

Capstone Project

Python Project Submission - Rahul

Movie Suggestion App using Py

Project Description

The Movie Suggestion App is a Python code that recommends movies to users based on their preferred genre and keywords.

The project integrates web scraping, data processing, and recommendation logic using real-world movie data sourced from IMDb.

The application first scrapes movie data from the IMDb Top 250 Movies chart, processes and stores the data, and then provides movie suggestion using based on user input.

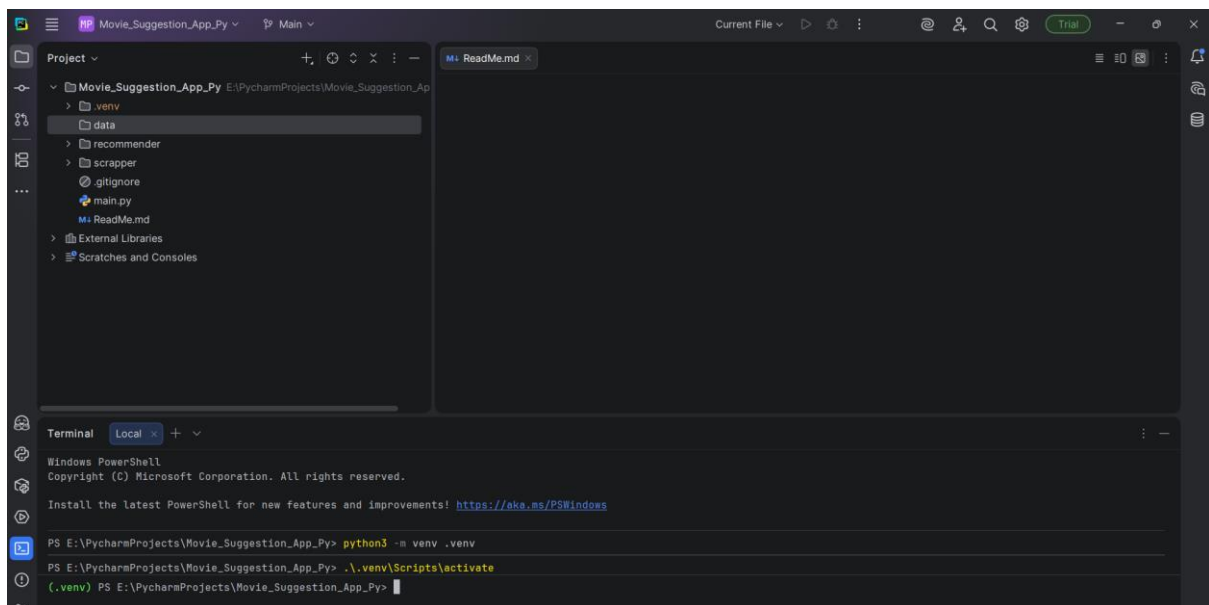
GITHUB: https://github.com/xrahulcrx/Python_Project_Submission-Rahul

How to Run the Project

Create & Activate Virtual Environment

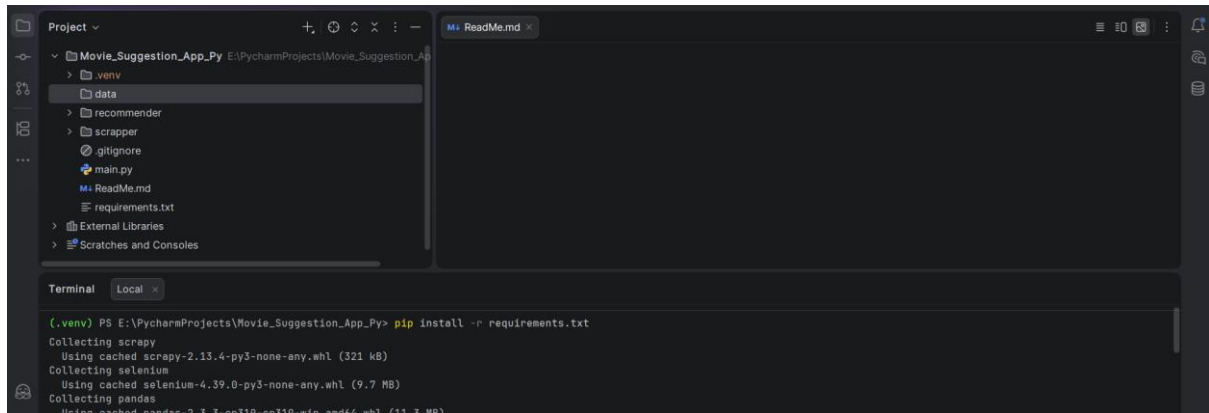
```
python -m venv .venv
```

```
.venv\Scripts\activate
```



Install Required Libraries

`pip install -r requirements.txt`



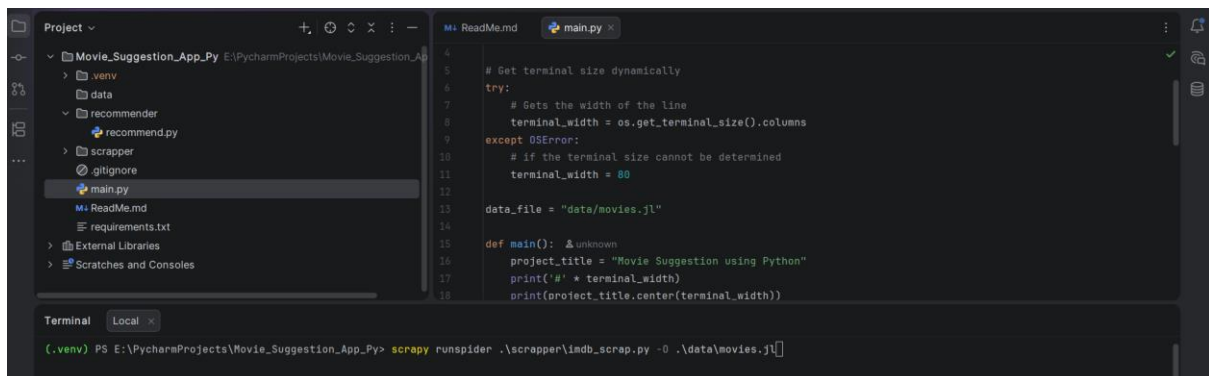
The screenshot shows the PyCharm IDE interface. The left sidebar displays the project structure for 'Movie_Suggestion_App_Py', including folders like '.venv', 'data', 'recommender', 'scraper', and files like 'main.py', 'ReadMe.md', and 'requirements.txt'. The main editor window shows the 'ReadMe.md' file. The bottom terminal window displays the command `pip install -r requirements.txt` and its output, which includes the collection of 'scrapy', 'selenium', and 'pandas' libraries using cached wheels.

Run the Scraper

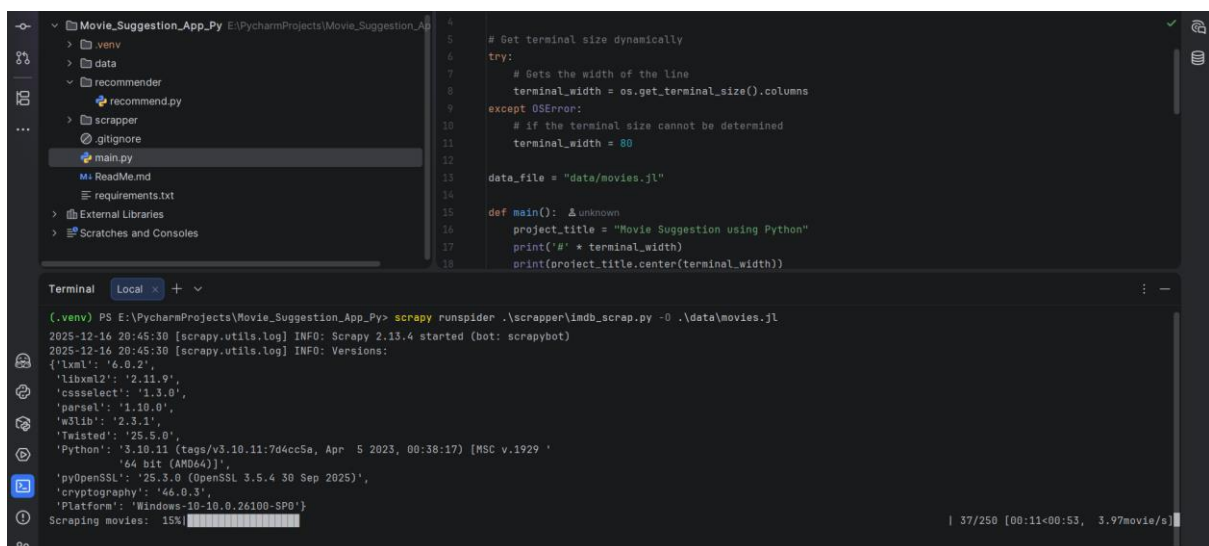
-O ensures the file is overwritten each time instead of appending.

`python -m scrapy runspider scraper/imdb_scrap.py -O data/movies.json`

When running the scraper, it will trigger the browser using selenium to fetch the data.



The screenshot shows the PyCharm IDE with the 'main.py' file open in the editor. The code defines a terminal width and a data file path, and includes a main function that prints the project title. The terminal window shows the command `scrapy runspider .\scraper\imdb_scrap.py -O .\data\movies.json` being executed.



The screenshot shows the PyCharm IDE with the 'main.py' file open in the editor. The terminal window displays the output of the scraper, including the scrapy version (2.13.4) and the versions of various dependencies like lxml, libxml2, cssselect, parsel, w3lib, Twisted, Python, and pyOpenSSL. The terminal also shows the progress of scraping movies, with a progress bar indicating 15% completion.

```
4
5 # Get terminal size dynamically
6 try:
7     # Gets the width of the line
8     terminal_width = os.get_terminal_size().columns
9 except OSError:
10    # if the terminal size cannot be determined
11    terminal_width = 80
12
13 data_file = "data/movies.json"
14
15 def main():
16     project_title = "Movie Suggestion using Python"
17     print('#' * terminal_width)
18     print(project_title.center(terminal_width))
```

Terminal Local + -

```
(.venv) PS E:\PycharmProjects\Movie_Suggestion_App_Py> scrapy runspider .\scraper\imdb_scrap.py -o .\data\movies.json
2025-12-16 20:45:30 [scrapy.utils.log] INFO: Scrapy 2.13.4 started (bot: scrapybot)
2025-12-16 20:45:30 [scrapy.utils.log] INFO: Versions:
{'lxml': '4.0.2',
 'libxml2': '2.11.9',
 'cssselect': '1.3.0',
 'parsel': '1.10.0',
 'w3lib': '2.3.1',
 'Twisted': '25.5.0',
 'Python': '3.10.11 (tags/v3.10.11:7d4cc5e, Apr 5 2023, 00:38:17) [MSC v.1929 '
 '64 bit (AMD64)]',
 'pyOpenSSL': '25.3.0 (OpenSSL 3.5.4 30 Sep 2025)',
 'cryptography': '46.0.3',
 'Platform': 'Windows-10-10.0.26100-SP0'}
Scraping movies: 100% | 250/250 [01:03:00:00, 3.91movie/s]
```

Completion of scrapper.

Run the Movie Suggestion App

python main.py

Sample Output:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS>python3 .\main.py
Movie Suggestion using Python

Enter genre (Drama, Action, Crime, etc.): drama
Enter keyword (optional - press Enter to skip): prison

Recommended Top 3 Movies:

The Shawshank Redemption (1994)
Rating: 9.3
Genre: Drama
Plot:A banker convicted of uxoricide forms a friendship over a quarter century with a hardened convict, while maintaining his innocence and trying to remain hopeful through simple compassion.
-----
The Green Mile (1999)
Rating: 8.6
Genre: Crime, Drama, Fantasy
Plot:A death row guard learns that a gentle giant in his charge possesses a mysterious gift.
-----
American History X (1998)
Rating: 8.5
Genre: Crime, Drama
Plot:A white supremacist is released from prison a changed man and hopes to prevent his younger brother from following in his footsteps.
-----
(.venv) PS E:\PycharmProjects\Python_Project_Submission-Rahul-Main>
```